

# UbuntuRemit: ASCO Research and Feasibility Paper

## 1. Executive Summary

This document outlines the foundational research and feasibility analysis for the Agentic Settlement and Compliance Orchestrator (ASCO) under the UbuntuRemit project. It evaluates the regulatory, technical, and market constraints of executing cross-border remittances within the African continent in 2026. The research prioritizes the South African Reserve Bank (SARB) mandates, the African Continental Free Trade Area (AfCFTA) frameworks, and the global migration to ISO 20022 messaging.

## 2. Regulatory Landscape and Ecosystems

### 2.1 The African Continental Free Trade Area (AfCFTA)

The AfCFTA aims to create a single market for goods and services across 54 African nations. A critical blocker to this vision has been the friction of cross-border payments, which traditionally route through correspondent banks outside the continent. The introduction of the Pan-African Payment and Settlement System (PAPSS) allows for instant cross-border payments in local currencies. ASCO must be designed to interface with PAPSS APIs, requiring strict adherence to real-time liquidity reporting and regional AML (Anti-Money Laundering) checks.

### 2.2 South African Reserve Bank (SARB) Directives

In the wake of South Africa's intense focus on maintaining compliance after historical FATF

grey-listing concerns, SARB has accelerated its Payments Ecosystem Modernisation (PEM) program. This program mandates:

- **Absolute Traceability:** Every node in a transaction life-cycle must be identifiable and auditable.
- **Real-Time Gross Settlement (RTGS) Upgrades:** Systems must be capable of processing high-value transactions instantaneously without batch delays.
- **Strict KYC/AML Enforcement:** Any AI-driven routing must embed regulatory checks before execution, not retroactively.

### 3. The ISO 20022 Messaging Standard

By 2026, the global financial system has largely completed the migration from legacy MT (text-based) messages to ISO 20022 MX (XML-based) messages. This standard provides significantly richer data capabilities.

| Message Type | Function                   | ASCO Implementation Need                                                                   |
|--------------|----------------------------|--------------------------------------------------------------------------------------------|
| pacs.008     | Customer Credit Transfer   | Core payload for moving funds between retail or corporate entities across borders.         |
| pain.001     | Customer Initiation        | The initial request structured by the user-facing application before agent validation.     |
| camt.053     | Bank to Customer Statement | Utilized for reconciliation and providing the audit trail back to the core banking system. |

## 4. Hardware and Technical Feasibility

The proposed multi-agent architecture relies on running concurrent Large Language Models (LLMs) to handle complex reasoning. Utilizing the AMD AI Developer Cloud provides a highly feasible technical path.

- **Compute Constraints:** The AMD Instinct MI300X features 192GB of HBM3 memory. This allows for the simultaneous hosting of a 70B parameter model (for complex compliance reasoning) and a 32B or smaller model (for rapid liquidity calculations) on a single node.
- **Software Stack:** AMD ROCm 7.0 provides native support for modern inference engines like vLLM. This ensures that the system can utilize high-throughput serving without relying on proprietary CUDA ecosystems.
- **Data Privacy:** By running these models on dedicated, sovereign infrastructure rather than calling public cloud APIs, financial institutions can maintain compliance with data residency laws.

## 5. Conclusion

The ASCO module of UbuntuRemit is highly feasible. The convergence of available AMD MI300X compute, the urgency of SARB compliance standards, and the data-rich nature of ISO 20022 create a perfect environment for an agentic orchestration layer. The primary challenge will be ensuring the deterministic execution of the agent negotiation to satisfy financial auditors.